13/10/26

$\frac{d\underline{x}}{dt} = A\underline{x}$ — $n \times n$ matrix $\qquad \underline{x} = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \end{pmatrix}$

Activity 4.12

$A = \begin{pmatrix} -3 & \sqrt{2} \\ \sqrt{2} & -2 \end{pmatrix}$ $\qquad r_1 = -4 \quad r_2 = -1 \quad |A - r\underline{I}| = 0$

$\underline{v}^{(1)} = \begin{pmatrix} -\sqrt{2} \\ 1 \end{pmatrix} \qquad \underline{v}^{(2)} = \begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix}$

$A\underline{x} = r\underline{x}$

General sol^n $\quad \underline{x}(t) = c_1 \begin{pmatrix} -\sqrt{2} \\ 1 \end{pmatrix} e^{-4t} + c_2 \begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix} e^{-t} \qquad \forall t$

Phase portrait

(i) Consider $\underline{x}(t) = c_2 \begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix} e^{-t} \quad (c_1 = 0)$

either $x_1 = c_2 e^{-t} \qquad x_2 = \sqrt{2} c_2 e^{-t} = \sqrt{2} x_1$ straight line gradient $\sqrt{2}$.

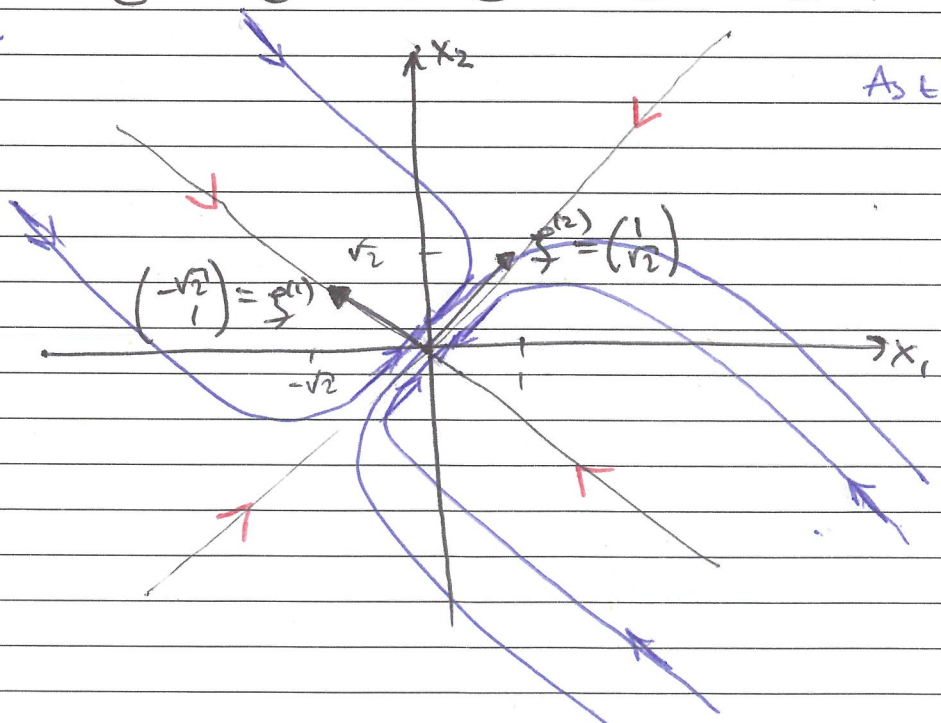
(ii) Consider $\underline{x}(t) = c_1 \begin{pmatrix} -\sqrt{2} \\ 1 \end{pmatrix} e^{-4t} \quad (c_2 = 0)$

$x_1 = -\sqrt{2} c_1 e^{-4t} \qquad x_2 = c_1 e^{-4t} = \frac{x_1}{-\sqrt{2}}$ straight line gradient $-\frac{1}{\sqrt{2}}$

(iii) $\underline{x}(t) = c_1 \begin{pmatrix} -\sqrt{2} \\ 1 \end{pmatrix} e^{-4t} + c_2 \begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix} e^{-t}$

As $t \to \infty$ near origin $\quad e^{4t} \gg e^{t} \quad \therefore \; e^{-4t} \ll e^{-t} \qquad \underline{x}(t) \to c_2 \begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix} e^{-t}$

As $t \to -\infty$ $\underline{x}(t) \to \begin{pmatrix} -\sqrt{2} \\ 1 \end{pmatrix}$



Real eigenvalues of the same sign : **node**

Complex conjugate eigenvalues

$$(A - r_1 I)\underline{\xi}^{(1)} = 0 \qquad \text{evalue } r_1, \text{ evector } \underline{\xi}^{(1)} \qquad (4.20)$$

Take complex conjugate:

$$(\bar{A} - \bar{r}_1 \bar{I})\bar{\underline{\xi}}^{(1)} = 0$$

$$(A - \bar{r}_1 I)\bar{\underline{\xi}}^{(1)} = 0 \quad (4.21)$$

$\Rightarrow \bar{r}_1, \bar{\underline{\xi}}^{(1)}$ are eigenvalue eigenvector pair

Example 4.13 $\quad \dfrac{d\underline{x}}{dt} = \begin{pmatrix} -1/2 & 1 \\ -1 & -1/2 \end{pmatrix}\underline{x}$

Eigenvalues: $|A - rI| = 0 \Rightarrow \begin{vmatrix} -1/2 - r & 1 \\ -1 & -1/2 - r \end{vmatrix} = 0 \qquad I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$$(1/2 + r)^2 + 1 = 0 \quad \Rightarrow r_{1,2} = -\tfrac{1}{2} \pm i \quad \text{complex conjugate pair}$$

Eigenvectors:

(i) $r_1 = -\frac{1}{2} + i$

$$(A - r_1 I)\underline{\xi}^{(1)} = \underline{0} \quad \Rightarrow \begin{pmatrix} -1/2 - (-1/2 + i) & 1 \\ -1 & -1/2 - (-1/2 + i) \end{pmatrix}\begin{pmatrix} \xi_1^{(1)} \\ \xi_2^{(1)} \end{pmatrix} = \underline{0}$$

$$\left.\begin{aligned} -i\,\xi_1^{(1)} + \xi_2^{(1)} &= 0 \\ -\xi_1^{(1)} - i\,\xi_2^{(1)} &= 0 \end{aligned}\right\} \text{ Take } \underline{\xi}^{(1)} = \begin{pmatrix} 1 \\ i \end{pmatrix} \qquad \left(A\underline{\xi}^{(1)} = r_1\underline{\xi}^{(1)}\right)$$

(ii) $r_2 = -\frac{1}{2} - i \qquad \underline{\xi}^{(2)} = \begin{pmatrix} 1 \\ -i \end{pmatrix}$

A fundamental set of solutions is $(\omega \neq 0)$

$$\underline{x}^{(1)}(t) = \begin{pmatrix} 1 \\ i \end{pmatrix} e^{(-\frac{1}{2}+i)t} \qquad \underline{x}^{(2)} = \begin{pmatrix} 1 \\ -i \end{pmatrix} e^{(-\frac{1}{2}-i)t}$$

$$\underline{x}(t) = c_1\begin{pmatrix} 1 \\ i \end{pmatrix} e^{(-\frac{1}{2}+i)t} + c_2\begin{pmatrix} 1 \\ -i \end{pmatrix} e^{(-\frac{1}{2}-i)t} \;//$$

To find the real-valued solutions, we need the real and imaginary parts of $\underline{x}^{(1)}$ (or $\underline{x}^{(2)}$)

$$\underline{x}^{(1)} = \begin{pmatrix} 1 \\ i \end{pmatrix} e^{-\frac{1}{2}t} e^{it}$$

$$= \begin{pmatrix} 1 \\ i \end{pmatrix} e^{-\frac{1}{2}t}(\cos t + i \sin t)$$

$$= e^{-\frac{1}{2}t}\begin{pmatrix} \cos t + i \sin t \\ i\cos t - \sin t \end{pmatrix}$$

$$= \underbrace{e^{-\frac{1}{2}t}\begin{pmatrix} \cos t \\ -\sin t \end{pmatrix}}_{\underline{m}(t)} + i\underbrace{e^{-\frac{1}{2}t}\begin{pmatrix} \sin t \\ \cos t \end{pmatrix}}_{\underline{n}(t)} \qquad \text{see (4-23)}$$

$\underline{m}(t) = e^{-1/2t}\begin{pmatrix}\cos t\\ -\sin t\end{pmatrix}$ and $\underline{n} = e^{-1/2t}\begin{pmatrix}\sin t\\ \cos t\end{pmatrix}$ is a set of real-valued solutions. [check $\frac{d}{dt}\underline{m} = A\underline{m}$ and similarly for $\underline{n}$]

Wronskian $W[\underline{m},\underline{n}] = \begin{vmatrix} e^{-1/2t}\cos t & e^{-1/2t}\sin t \\ -e^{-1/2t}\sin t & e^{-1/2t}\cos t\end{vmatrix}$

$= e^{-t}\cos^2 t + e^{-t}\sin^2 t$

$= e^{-t} \neq 0 \quad \therefore \underline{m}$ and $\underline{n}$ are a fundamental set

The general solution (th^m 4.10) is $\underline{x}(t) = c_1 e^{-1/2t}\begin{pmatrix}\cos t\\ -\sin t\end{pmatrix} + c_2 e^{-1/2t}\begin{pmatrix}\sin t\\ \cos t\end{pmatrix}$, $\forall t$

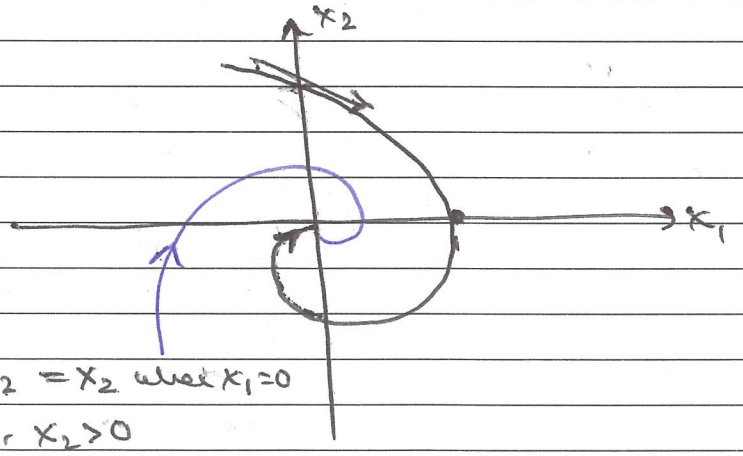
Phase portrait

(1) Consider $\underline{x}(t) = c_1 e^{-1/2t}\begin{pmatrix}\cos t\\ -\sin t\end{pmatrix}$ $(c_2 = 0)$

$x_1 = c_1 e^{-1/2t}\cos t \qquad x_2 = -c_1 e^{-1/2t}\sin t$

$x_1^2 + x_2^2 = c_1^2 e^{-t}$ "circle of decreasing radius" spiral

$\underline{x}^{(1)}(t=0) = c_1 e^{0}\begin{pmatrix}\cos 0\\ -\sin 0\end{pmatrix} = c_1\begin{pmatrix}1\\ 0\end{pmatrix}$



$\frac{dx_1}{dt} = -\frac{1}{2}x_1 + x_2 = x_2$ when $x_1 = 0$

$\frac{dx_1}{dt} > 0$ for $x_2 > 0$

- Complex eigenvalues with negative real part: origin is a spiral "asymptotically stable".
- C.C evalues with positive real part: spiral unstable.
- Purely imagnary eigenvalues: centre